Fiscal Year:	FY 2008	**Task Last Updated:**	FY 08/12/2008
PI Name:	Globus, Ruth Ph.D.		
Project Title:	Simulated Microgravity and Radiation-Induced Bone Degeneration: Oxidative Stress- and p53-Dependent Mechanisms		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation Biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	(1) **Bone Fracture**:Risk of Bone Fracture due to Spaceflight-induced Changes to Bone (2) **Cardiovascular**:Risk of Cardiovascular Adaptations Contributing to Adverse Mission Performance and Health Outcomes (3) **Osteo**:Risk Of Early Onset Osteoporosis Due To Spaceflight		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	Ruth.K.Globus@nasa.gov	**Fax:**	FY
PI Organization Type:	NASA CENTER	**Phone:**	650-604-5247
Organization Name:	NASA Ames Research Center		
PI Address 1:	Bone and Signaling Laboratory		
PI Address 2:	Space Biosciences Research Branch		
PI Web Page:			
City:	Moffett Field	**State:**	CA
Zip Code:	94035-1000	**Congressional District:**	18
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2004 Radiation Biology NNH04ZUU005N
Start Date:	10/01/2005	**End Date:**	03/31/2010
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	3	**Monitoring Center:**	NASA ARC
Contact Monitor:	Cucinott1a, Francis	**Contact Phone:**	281-483-0968
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Flight Program:			
Flight Assignment:	NOTE: Received NCE to 3/31/2010 per A. Chu/ARC (7/2009) NOTE: Changed Division and Discipline/Program to HRP as of FY2006, per program changes at that time, per JSC/A. Chu-ARC (jvp 4/2009)		
Key Personnel Changes/Previous PI:	Dr. Kenji Yumoto is now the lead postdoctoral fellow on this projects, as Dr. Hisataka Kondo accepted the offer of a faculty position.		
COI Name (Institution):	Almeida, Eduardo A.C. (NASA Ames Research Center) Limoli, Charles (University of California Irvine) Vercoutere, Wenonah (NASA Ames Research Center) Searby, Nancy (NASA Ames Research Center)		
Grant/Contract No.:	INTRAMURAL		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	To determine the risks and mechanisms of tissue degeneration in the spaceflight environment, we will examine the combined effects of radiation and simulated weightlessness on bone. We will focus our studies on two important pathways, p53 and oxidative and nitrative stress-related species, in bone marrow progenitor populations because there is abundant evidence that the biological effects of weightlessness and those of radiation share these mechanism. Experiments will be conducted using the rodent hindlimb unloading model, the best validated animal model for simulating weightlessness, followed by irradiation. In Aim 1, we will test the hypothesis that simulated weightlessness of mice modulates the response of bone and bone marrow to radiation, both acutely and during tissue regeneration. In Aim 2 we will identify the target marrow-derived stem, progenitor and differentiated cell populations sensitive to the combined stimuli of simulated weightlessness and radiation. In Aim 3, we will explore the relevance of specific ROS/RNS and p53-dependent mechanisms for mediating cellular responses to combined simulated weightlessness and radiation. The results of these studies are expected to provide important information for the development of countermeasures both to help prevent degeneration of connective tissues such as bone during spaceflight and to improve subsequent tissue regeneration.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This project addresses several of the risks identified in NASA's Bioastronautics Critical Path Roadmap (BCPR) caused by long duration spaceflight. We will obtain basic new knowledge about bone tissue degeneration caused by musculoskeletal disuse and radiation at the molecular, cell, and tissue level, relevant to Earth, as well as in space. Results will also lead to the formulation of new countermeasures concepts and risk assessment strategies for bone and connective tissue degenerative diseases.
Task Progress:	Our previous studies demonstrated that gamma irradiation and hypergravity caused oxidative stress in the form of elevated reactive oxygen species (ROS) in bone marrow cells. Gamma irradiation also caused acute loss of cancellous bone, which is a metabolically active tissue. Our prior results showed that redox sensitive pathways activated by gravity and irradiation cause key physiologic changes in skeletal cells. In this last year, we performed two series of experiments at NASA Space Radiation Laboratory-Brookhaven National Laboratory (using the heavy particle, Fe-56), and separate experiments at NASA Ames Research Center (using the gamma emitter, Cs-137). Analysis of the first set of experiments, which focused on the acute effects of Fe-56 with musculoskeletal disuse, is now nearing completion. Analysis of samples from the second set of experiments conducted at BNL is still in progress. We found heavy particle radiaiton (Fe-56), which is relevant to space radiation, at relatively low doses appeared to cause rapid (within 3 days) cancellous bone loss. Acute bone loss appeared maximal at the lowest dose of heavy ion radiation we tested (10 cGy), although the effects are relatively small and longer duration experiments are needed to confirm this result. To determine the influence and dose-dependence of heavy particle and gamma irradiation on the marrow cells that mature to become bone-forming osteoblasts bone resorbing osteoclasts, bone marrow cells were irradiated with Fe-56 or Cs-137 while growing under suitable culture conditions to stimulate maturiation. We found that irradiation of cultured marrow cells inhibited maturation of both types of skeletal cells, despite the finding that in the intact tissue, irradiation increased the breakdown of bone. These results indicate that systemic or local factors within the tissue are likely to play a key role in causing bone loss triggered by irradiation. A model that simulates weightlessness was used to determine if musculoskeletal disuse impacts sensitivity of osteoblasts to irradiation and acute bone loss. We found that both simulated weightlessness and irradiation both inhibited osteoblast maturation and in combination, the effects of simulated weightlessness and irradiation on osteoblast maturation were even worse. Based on these results we hypothesized disuse sensitized bone marrow cells to irradiation and predicted that radiation exposure in microgravity can impair skeletal recovery from simulated weightlessness. We are now analyzing samples from a second set of studies performed at NSRL/BNL that were designed to determine whether or not the adverse effects of heavy particle irradiation on bone marrow cells persist and can prevent the normal recovery of skeletal tissue from disuse. In separate experiments conducted at NASA ARC, we found that the detrimental effects of gamma irradiation (2 Gy) on cancellous bone microarchitecture persisted one month after irradiation, although irradiation did not affect more dense cortical bone or biomechanical properties of lumbar vertebrae. Thus, gamma irradiation alone did not adversely affect biomechanical properties of vertebrae one month after exposure despite the observed bone loss, which we attribute to the ability of dense, cortical bone to preserve the spine's strength. In humans, a relatively small loss of bone in cancellous tissue can lead to large changes in bone strength. Persistent cancellous bone loss, as observed in these experiments caused by irradiation, may lead to increase fragility at a later time, perhaps during aging, bedrest or spaceflight. Our results have potential clinical relevance to astronauts as well as to cancer patients treated therapeutically with radiation.
Bibliography Type:	Description: (Last Updated: 06/04/2025)
Abstracts for Journals and Proceedings	Alwood JS, Yumoto Y, Kondo H, Mojarrab R, Lindsey DP, Beaupré GS, Limoli CL, Almeida EAC, Searby ND, Globus RK. "Combined Cs-137 gamma irradiation and hindlimb unloading rapidly decreased cancellous bone volume faction in the mouse lumbar vertebrae." American Society for Bone and Mineral Metabolism Annual Meeting, Montreal, Sept 2-16, 2008. Journal of Bone and Mineral Research, Suppl., 2008. , Sep-2008
Abstracts for Journals and Proceedings	Yumoto Y, Mojarrab R, Arakaki J, Wang A, Hilton D, Almeida EAC, Limoli C, Searby ND, Globus RK. "Musculoskeletal disuse worsens the acute detrimental effects of heavy particle radiation on osteoblastogenesis." Presented at the American Society for Bone and Mineral Metabolism, Meeting in Montreal, Montreal, Canada, Sept 12-16, 2008. Journal of Bone and Mineral Research, Suppl., 2008. , Sep-2008

Abstracts for Journals and Proceedings	Yumoto K, Mojarrab R, Arakaki J, Wang A, Hilton D, Almeida EEAC, Limoli C, Searby ND, Globus RK. "Adverse effects of Fe-56 irradiation and simulated weightlessness on osteoblastogenesis." NASA Radiation Program PI meeting , Philadelphia, PA, July 2008. NASA Radiation Program PI meeting , Philadelphia, PA, July 2008. , Jul-2008
Abstracts for Journals and Proceedings	Globus RK, Yumoto K, Kondo H, Alwood J, Mojarrab R, Wang A, Hilton D, Almeida EAC, Searby ND, Limoli C. "Simulated weightlessness and radiation-induced bone loss in mice." Presented at NASA Human Research Program Principal Investigators' Workshop, League City, TX, February 2008. NASA Human Research Program Principal Investigators' Workshop, League City, TX, February 2008. , Feb-2008
Awards	Yumoto K, Mojarrab R, Arakaki J, Wang A, Hilton D, Almeida EEAC, Limoli C, Searby ND, Globus RK. "1st place award for student science presentation, '"Adverse effects of Fe-56 irradiation and simulated weightlessness on osteoblastogenesis.' NASA Radiation Program PI meeting , Philadelphia, PA, July 2008." Jul-2008